POLARIS Stage 8.2 — Data Source and Observability Breakthrough Research Report

1. OBSERVABILITY GAP

The POLARIS Stage 8.1 operational champion model operates on a 37-feature predictive architecture. It integrates historical surface weather from the Swedish Infrastructure for Ecosystem Science (SITES) Nuolja station, clear-sky solar geometry, 3D vertical liquid and ice water path integrations from the MetCoOp Ensemble Prediction System (MEPS), 360-degree local terrain horizon profiles derived from 30m Copernicus DEM data, and 700 hPa free-atmosphere steering wind vectors (U_{700}, V_{700}). On the frozen held-out out-of-distribution (ODD) test set comprising 173 daily forecast origins, Stage 8.1 achieved a daytime Mean Absolute Error (MAE) of 36.1887 \text{ W/m}^2, a daytime normalized MAE (nMAE) of 28.0688\%, and a Root Mean Square Error (RMSE) of 64.2195 \text{ W/m}^2. Although a 10,000-resample paired bootstrap indicated a 96.65\% probability of candidate improvement over Stage 7.5, the 95% confidence interval [-0.8880, +0.0264] \text{ W/m}^2 spans zero. This statistical ceiling indicates that transforming gridded Numerical Weather Prediction (NWP) parameters has reached an informational asymptote.
To break through the 28% nMAE threshold, POLARIS must target the unobserved physical properties of the polar atmosphere above Nuolja Middle Station (68.362074^\circ\text{N}, 18.738752^\circ\text{E}, elevation 675 \text{ m a.s.l.}). The remaining forecast error is driven by three primary observability gaps:

Real-Time Upstream Cloud Optical Thickness (\tau) and Droplet Microphysics

Operational NWP models calculate surface shortwave irradiance (SW_IN) by passing parameterized cloud water (q_l) and cloud ice (q_i) through radiative transfer schemes such as RRTMG. In Arctic and sub-Arctic alpine environments, NWP microphysical parameterizations frequently miscalculate cloud droplet number concentration, effective radius (r_e), liquid-to-ice phase transitions, and sub-grid optical extinction coefficients. POLARIS currently observes no physical measurements of cloud optical depth (\tau), cloud droplet effective radius (r_e), or liquid/ice phase partitioning prior to forecast origin t_0. Instead, the model relies on model-simulated cloud paths or historical local ground-variance proxies such as ctx_std_6h.

Real-Time Upstream Spatial Cloud Structure and Observed Cloud Displacement

While MEPS supplies gridded cloud fraction and 700 hPa steering winds (U_{700}, V_{700}), POLARIS lacks real-time observations of the actual spatial cloud fields surrounding Abisko and Nuolja prior to t_0. The model cannot observe incoming cloud boundaries, spatial cloud heterogeneity, or empirically tracked Cloud Motion Vectors (CMVs). When MEPS misjudges the propagation speed, spatial extent, or arrival time of an upstream cloud front moving across the Scandinavian mountains, POLARIS lacks independent observational data to correct the timing

offset.

Partitioned Surface Radiation Components and Boundary-Layer Turbidity

Ground-truth observations at SITES Nuolja are limited to unpartitioned incoming global shortwave radiation (SW_IN). POLARIS does not observe the physical split between Direct Normal Irradiance (DNI) and Diffuse Horizontal Irradiance (DHI). Under thin cirrus, mixed-phase stratus, or broken cloud regimes, radiative transfer is governed by single-scattering albedo and atmospheric turbidity. POLARIS has no direct observations of atmospheric aerosol optical depth, boundary-layer aerosol loading, or real-time ground-level diffuse fractions.

2. TOP 15 DATA SOURCES

Fourteen independent remote-sensing, ground-based, and atmospheric data sources covering 68.362074^\circ\text{N}, 18.738752^\circ\text{E} were evaluated to address these observability gaps. Every data-availability claim is categorized under strict archive verification standards (VERIFIED, LIKELY, UNCERTAIN, or NOT VERIFIED).

Rank	Data Source & Provider	Targeted Product	Atmospheric Variables Observed	Spatial & Temporal Resolution	2024 Nuolja Coverage & Overpass Yield	Causal Status (t \le t_0)	Archive Verification Status	Extraction Burden	Physical Novelty & Expected Usefulness
1	**NASA / NOAA Earthdata**	CLDPROP_L2_VIIRS_SNPP & NOAA20	Cloud Optical Thickness (\tau), Effective Radius (r_e), Cloud Phase	750 m swath / 14–28 passes daily	High-latitude polar orbit overpasses covering 68.36°N	Causal (t_{\text[span_1](start_span)[span_1](end_span)[span_15](start_span)[span_15](end_span){obs}} \le t_0, Latency: ~3 h)	**VERIFIED**	Moderate (LAADS DAAC NetCDF 4 API)	Direct satellite measurement of cloud optical extinction and droplet radius
2	**Copernicus / ESA**	S5P_OFFL_L2__CLOUD / NRTI	Cloud Optical Thickness (\tau), Cloud Top Height, Cloud	3.5 km × 5.5 km / ~14 orbits daily	High-latitude polar overpasses covering 68.36°N	Causal (NRTI latency < 3 h)	**VERIFIED**	Moderate (Copernicus CDSE STAC API)	Independent oxygen A-band photon-path cloud depth retrieval

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			Fraction						
3	**EUMETSAT / MET Norway**	NWC SAF PPS Cloud Microphysics (CPP)	Cloud Optical Depth (\tau), Liquid Water Path (LWP), Cloud Phase	1 km to 4 km grid / 14–28 passes daily	Full regional Arctic coverage including Nuolja	Causal (t_{\text{obs}} \le t_0, Latency: 30–60 min)	**VERIFIED**	High (EUMETSAT Data Center / THREDDS)	High-latitude optimized satellite cloud property retrievals
4	**SITES Infrastructure**	Abisko Research Station (ANS) Main Base	Direct Irradiance (SW_DIR), Diffuse Irradiance (SW_DIFF), Ceilometer CBH	Point observation / 1-min to 10-min frequency	Continuous ground station ~3.5 km from Nuolja	Fully Causal (t_{\text{obs}} \le t_0, Latency < 5 min)	**VERIFIED**	Low (SITES Data Portal CSV download)	Ground-observed physical split of direct extinction vs. scattering
5	**EUMETSAT / NWC SAF**	PPS High Resolution Winds (HRW / CMV)	Cloud Motion Vectors (Speed, Direction, Height)	10–50 km vector grid / Polar overpass pairs	Arctic coverage via MetOp-B/C & NOAA pairs	Causal (t_{\text{obs}} \le t_0, Latency < 90 min)	**LIKELY**	High (NWC SAF polar processing software)	Directly observed physical cloud advection velocities
6	**SMHI Open Data**	Swedish Ceilometer Network (Kiruna/ Abisko)	Cloud Base Height (CBH), Multi-layer Cloud Structure	Point observation / 15-min reporting frequency	Regional coverage near Abisko region	Fully Causal (t_{\text{obs}} \le t_0, Real-time REST API)	**LIKELY**	Low (SMHI Open Data REST API)	Ground-observed boundary-layer cloud vertical structure
7	**Copernicus /**	cams-europe-air-	Aerosol Optical	0.1° × 0.1° (~10	Complete	Causal (t_{\text{r	**VERIFIED**	Moderate	Dynamic optical

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	CAMS	quality-forecasts	Depth at 550nm (aod550), Dust AOD	km) / Hourly analysis frequency	gridded coverage over Abisko region	ef}} < t_0, Initialized 00Z/12Z)		(Copernicus Atmosphere Data Store)	extinction from atmospheric aerosols
8	**EUMETSAT / CM SAF**	SARAH-3 / CLARA-A3 Polar Cloud Record	Surface Shortwave Radiation, Cloud Optical Thickness	0.05° × 0.05° (~5 km) / Hourly & daily frequency	Full Nordic region coverage	Non-Causal for NRT (ICDR latency > 24–48 h)	**LIKELY**	Moderate (EUMETSAT CM SAF User Interface)	Historical reference dataset; non-causal for real-time operations
9	**Copernicus / ESA**	Sentinel-3 SLSTR Level-2 Cloud (SL_2_LST)	Cloud Fraction, Cloud Top Temperature, SWIR Radiance	500 m SWIR / 1 km Thermal / ~2 passes daily	Polar orbit overpasses over 68.36°N	Causal (t_{\text{obs}} \le t_0, Latency ~3 h)	**VERIFIED**	High (Copernicus CDSE STAC API)	High-resolution spatial cloud masks and top temperatures
10	**SMHI / MET Norway**	BALTRAD Nordic Composite Radar Network	Composite Radar Reflectivity (dBZ), Echo Tops	1 km grid / 15-min composite frequency	Partial coverage (Kiruna radar beam offset by topography)	Causal (t_{\text{obs}} \le t_0, Latency < 15 min)	**LIKELY**	High (SMHI BALTRAD node)	Detection of precipitating mixed-phase cloud cores
11	**EUMETSAT / AC SAF**	GOME-2 / IASI Multi-Sensor Aerosol (PMAO)	Aerosol Optical Depth, Absorbing Aerosol Index	12 km to 40 km swath / Daily polar passes	High-latitude polar orbit tracks	Causal (t_{\text{obs}} \le t_0, Latency ~3 h)	**UNCERTAIN**	High (EUMETSAT Data Centre)	Polar atmospheric aerosol loading measurements
12	**Copernicus /**	CAMS Solar	Clear-sky GHI,	Point interpolat	Point extractio	Non-Causal for	**VERIFIED**	Low (CAMS	Reanalysis

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	CAMS	Radiation Service (CAMS-RAD)	Direct Irradiance, Water Vapor	ion / 15-min time step	n for Nuolja coordinates	NRT (Product latency ~24 h)		ADS API)	clear-sky GHI accounting for water vapor
13	**NOAA / NESDIS**	PATMOS-x Advanced Polar Cloud Product	Cloud Phase, Liquid Water Path, Cloud Top Height	4 km equal-area grid / Daily polar composite	Arctic regional grid coverage	Causal (t_{\text{obs}} \le t_0, Latency ~4–6 h)	**LIKELY**	High (NOAA CLASS Archive)	Historical polar cloud microphysics records
14	**NASA Langley**	CERES FLASHFlux Level-2 SSF	Cloud Optical Depth, Surface Direct & Diffuse Shortwave	20 km footprint / Overpass sampling	Polar orbits over 68.36°N	Non-Causal for NRT (Product latency ~3–4 days)	**VERIFIED**	High (NASA Langley Earthdata)	Satellite-derived surface radiation budget fields

3. TOP 5 BREAKTHROUGH SOURCES

The top five breakthrough data sources were selected for their physical relevance and ability to address specific observability gaps in the POLARIS forecasting pipeline.

1. NASA/NOAA VIIRS Level-2 Cloud Properties Continuity Product (CLDPROP_L2_VIIRS_SNPP and NOAA20)

POLARIS's previous thermal-IR satellite experiment yielded only 334 valid observations across 2024, primarily from legacy AVHRR sensors. That experiment relied on thermal brightness temperatures (T_b), which register cloud-top altitude but lack physical sensitivity to cloud optical thickness.

CLDPROP_L2 resolves this issue by utilizing bi-spectral solar reflectance—pairing the 0.67 \mu m visible channel with 1.6 \mu m and 2.1 \mu m shortwave-infrared channels—to retrieve Cloud Optical Thickness (\tau) and Cloud Droplet Effective Radius (r_e) at 750 m pixel resolution. At 68.36^\circ\text{N}, polar orbit convergence causes Suomi-NPP, NOAA-20, and NOAA-21 to cross the Abisko region between 14 and 28 times per day. This high overpass yield provides continuous spatial snapshots of Cloud Optical Thickness prior to t_0, giving the model a direct

physical measurement of cloud optical extinction before clouds advect over Nuolja.

2. Sentinel-5P TROPOMI Level-2 Cloud Product (S5P_OFFL_L2__CLOUD / S5P_NRTI_L2__CLOUD)

The TROPOMI instrument onboard Sentinel-5P measures high-resolution spectral absorption within the oxygen A-band (758–771 nm) using the FRESCO and ROCINN retrieval algorithms. TROPOMI provides independent physical estimates of Cloud Optical Thickness (cloud_optical_thickness), Cloud Top Height (cloud_top_height), and Cloud Base Height (cloud_base_height) at 3.5 \text{ km} \times 5.5 \te[span_48](start_span)[span_48](end_span)xt{ km} resolution.
While MEPS pressure-level vertical integrations provide model-simulated cloud boundaries, TROPOMI supplies an independent photon-path physical measurement of cloud vertical structure and optical depth. The Near-Real-Time (NRTI) processing stream is disseminated within 3 hours of overpass, satisfying the causal availability constraint (t_{\text{obs}} \le t_0).

3. EUMETSAT NWC SAF PPS Cloud Microphysics (CPP) Derived from MetOp-B/C

The EUMETSAT Satellite Application Facility on Support to Nowcasting (NWC SAF) Polar Processed Software (PPS) is tailored for high-latitude polar environments. The Cloud Physical Properties (CPP) module processes AVHRR-3 and VIIRS instrument feeds to generate Cloud Phase (liquid, ice, supercooled liquid), Cloud Liquid Water Path (CLWP), and Cloud Optical Depth (\tau).
POLARIS currently relies on MEPS parameters for cloud water integration. NWC SAF CPP provides observed microphysical cloud water content derived from regional direct-broadcast receiver nodes operated by SMHI and MET Norway. Disseminated within 30 to 60 minutes of overpass, it offers real-time microphysical data to correct MEPS liquid/ice distribution errors.

4. SITES Abisko Scientific Research Station (ANS) Main Base Network

SITES Nuolja Middle Station records unpartitioned incoming global shortwave radiation (SW_IN). The primary Abisko Scientific Research Station (~3.5 km away, elevation 385 m a.s.l.) maintains continuous measurements of Direct Normal Irradiance (SW_DIR), Diffuse Horizontal Irradiance (SW_DIFF), and ceilometer Cloud Base Height (CBH).
The direct-to-diffuse ratio (\text{DHI} / \text{GHI}) serves as an empirical indicator of single-scattering albedo and cloud extinction within the Abisko valley. Ingesting pre-forecast diffuse ratios and exact cloud base altitudes provides POLARIS with observed local atmospheric transmission statistics to complement regional NWP inputs.

5. EUMETSAT NWC SAF Polar High Resolution Winds (HRW / Cloud Motion Vectors)

Stage 8.1 incorporated 700 hPa NWP steering winds (U_{700}, V_{700}) to proxy cloud advection. NWC SAF HRW extracts actual Cloud Motion Vectors (CMVs) by tracking cloud tracer features across consecutive polar satellite passes (e.g., MetOp-B followed by MetOp-C). HRW outputs observed cloud drift velocity vectors (speed, direction, and target pressure altitude) across Scandinavian terrain. Ingesting observed cloud displacement vectors enables POLARIS to compute physical advection lag times for upstream satellite-observed cloud optical

thickness features.

4. ONE BEST NEW INFORMATION SOURCE

The single highest-priority data source recommended for Stage 8.2 is the **NASA/NOAA VIIRS Level-2 Cloud Properties Continuity Product (CLDPROP_L2_VIIRS_SNPP and CLDPROP_L2_VIIRS_NOAA20)**.

Physical Information Added

CLDPROP_L2 provides pixel-level observations of Cloud Optical Thickness (\tau, dimensionless, range 0–100), Cloud Droplet Effective Radius (r_e, micrometers), and Cloud Phase (Liquid, Ice, Supercooled Liquid) at 750 m spatial resolution.

Radiative Transfer Formulation

Solar shortwave radiation passing through an atmospheric cloud layer undergoes scattering and absorption governed by the radiative transfer equation:
I_{\text{surface}} = I_{\text{[span_5](start_span)[span_5](end_span)[span_19](start_span)[span_19](end_span)clearsky}} \cdot \exp\left(-\frac{\tau}{\cos \theta_z} \right) + I_{\text{diffuse}}(\tau, r_e, \varpi_0)
Where:

- \tau represents the satellite-observed Cloud Optical Thickness.
- r_e is the Cloud Droplet Effective Radius.
- \theta_z is the solar zenith angle.
- \va[span_6](start_span)[span_6](end_span)[span_20](start_span)[span_20](end_span)rpi_0 is the single-scattering albedo determined by cloud phase.

POLARIS currently infers \tau indirectly through NWP parameterized liquid water paths or ground historical fluctuations. CLDPROP_L2 supplies direct satellite measurements of \tau and r_e.
By defining a 50 \text{ km} \times 50 \text{ km} spatial Region of Interest (ROI) centered on Nuolja (68.362074^\cir[span_30](start_span)[span_30](end_span)c\text{N}, 18.738752^\circ\text{E}), POLARIS can extract four causal features prior to forecast origin t_0:

1. viirs_cot_mean_25km: Spatial mean Cloud Optical Thickness within 25 km of Nuolja.
2. viirs_cot_upstream_sector: Spatial mean Cloud Optical Thickness within a 45-degree upstream sector oriented along the Stage 8.1 700 hPa wind vector (U_{700}, V_{700}).
3. viirs_cot_heterogeneity: Spatial standard deviation of \tau, quantifying cloud patchiness.
4. viirs_cloud_phase_liquid_fraction: Percentage of cloud pixels classified as supercooled liquid water versus ice crystals.

5. FALSIFIABLE EXPERIMENT

To evaluate the predictive impact of VIIRS Cloud Optical Thickness without risking data leakage, benchmark modification, or ODD test contamination, the following experimental protocol is defined.

Protocol Locks

- **Target Variable**: SITES Nuolja 2024 hourly reconstructed incoming shortwave radiation (SW_IN, \text{W/m}[span_33](start_span)[span_33](end_span)^2).
- **Baseline Control**: Stage 8.1 Operational Champion (37 features: Stage 7.5 features + U_{700} + V_{700}).
- **Model Family & Hyperparameters**: HistGradientBoostingRegressor(loss="absolute_error", max_depth=4, min_samples_leaf=30, l2_regularization=10.0, random_state=42).
- **Dataset Partitioning**:
    - EVEN Origins (174 daily origins / 2,885 daytime validation hours): Forward-chaining 5-fold cross-validation.
    - ODD Origins (173 daily origins / 2,883 daytime validation hours): **STRICTLY LOCKED AND UNTOUCHED**.

Causal Ingestion Constraint

For every forecast origin t_0, feature extraction is restricted to VIIRS overpasses satisfying: t_{\text{overpass}} + \text{latency} \le t_0
Where \text{latency} = 3 \text{ hours} (conservative NRT processing and distribution window). If no valid overpass exists within t_0 - 6\text{h} \le t_{\text{overpass}} \le t_0 - 3\text{h}, the features default to clear-sky zero values with a missingness indicator flag (viirs_observation_valid = 0).

Predeclared EVEN Cross-Validation Acceptance Gates

1. **Gate 1 (Aggregate Daytime MAE)**: Candidate EVEN CV MAE \le 37.4201 \text{ W/m}^2 (\Delta \text{MAE} \le -0.20 \text{ W/m}^2 relative to Stage 8.1 control of 37.6201 \text{ W/m}^2).
2. **Gate 2 (Fold Consistency)**: Candidate must improve Daytime MAE in at least 4 of 5 forward-chaining CV folds.
3. **Gate 3 (Degradation Cap)**: Maximum single-fold degradation must not exceed +0.15 \text{ W/m}^2.
4. **Gate 4 (nMAE Bound)**: Candidate EVEN nMAE \le 37.9745\%.
5. **Gate 5 (RMSE Bound)**: Candidate EVEN RMSE \le 60.1606 \text{ W/m}^2.

6. DATA-AVAILABILITY VERIFICATION PLAN

Before developing feature extraction scripts, the engineering team must execute the following five-step archive verification procedure.

Step 1: Direct LAADS DAAC Archive Query

Query NASA LAADS DAAC for CLDPROP_L2_VIIRS_SNPP and CLDPROP_L2_VIIRS_NOAA20 granules covering bounding box [68.1^\circ\text{N}, 18.2^\circ\text{E}, 68.6^\circ\text{N}, 19.3^\circ\text{E}] for all 365 days of 2024. Document the exact granule count returned to confirm coverage completeness.

Step 2: Overpass Frequency and Gap Audit

Calculate the empirical distribution of temporal gaps between valid overpasses at 68.362074^\circ\text{N}, 18.738752^\circ\text{E}. Confirm that at least 85\% of daily t_0 forecast origins contain a valid VIIRS overpass within the preceding 6 hours.

Step 3: Latency Metadata Inspection

Extract metadata attributes GranuleBeginningDateTime and ProductionDateTime from 500 random 2024 NetCDF4/HDF5 granules. Verify that latency (\text{ProductionDateTime} - \text{GranuleBeginningDateTime}) is \le 3 \text{ [span_10](start_span)[span_10](end_span)[span_24](start_span)[span_24](end_span)hours} for at least 95\% of sample files to satisfy causal availability requirements.

Step 4: Variable Inspection & Quality Mask Audit

Inspect Science Data Sets (SDS) Cloud_Optical_Thickness and Cloud_Optical_Thickness_Quality_Flag using xarray. Confirm that missing data, nighttime pixels, and high-latitude low-sun pixels are explicitly masked using standard netCDF fill values (_FillValue = -9999.0 or 9.96921e36) without producing unmasked Infinities or NaNs.

Step 5: Spatial Co-Registration Grid Alignment

Perform spatial cross-registration between VIIRS pixel latitude/longitude arrays and Copernicus DEM 30m terrain grid points (y=760, x=486). Confirm that the geodetic distance between pixel centers and Nuolja station is \le 750 \text{ meters}.

7. DEAD ENDS

Five data sources and feature engineering directions were investigated and rejected due to physical, spatial, or causal limitations.

1. Geostationary Satellite Retrievals (Meteosat MSG SEVIRI)

Sub-satellite positioning for Meteosat Second Generation (MSG) is located at 0^\circ\text{N}, 0^\circ\text{E}. At Nuolja (68.36^\circ\text{N}), the Satellite Viewing Zenith Angle (VZA) exceeds 78^\circ–80^\circ. At these extreme slant angles, pixel footprints expand from 3 \text{ km} to >12 \text{ km}, causing parallax displacement, terrain shadowing, and path attenuation that invalidate cloud optical depth retrievals.

2. Global Atmospheric Reanalysis Radiation Datasets (ERA5 / MERRA-2)

ERA5 reanalysis fields are published with a 5-day to 2-month dissemination latency and assimilate observational data post-hoc. Ingesting ERA5 shortwave flux or cloud fields at forecast origin t_0 introduces future data leakage.

3. MEPS 18Z Native Model-Level Cloud Hydrometeors (q_l, q_i on 65 Levels)

As audited in Stage 8.0, MET Norway archives native 65-level model grid files (meps_det_ml) exclusively for the 00Z initialization run (0/347 availability at 18Z). Utilizing 00Z model-level

fields introduces an 18-hour forecast latency lag, making it unviable for 18Z operational runs.

4. Raw Uncalibrated Thermal-IR Brightness Temperatures (T_b)

Thermal-IR brightness temperature measures cloud-top temperature but lacks sensitivity to cloud optical thickness (\tau) or lower-level liquid water content in multi-layer Arctic cloud structures. Stage 6.2 evaluated thermal-IR point features and observed no out-of-sample performance improvement (\Delta \text{MAE} = +0.0290 \text{ W/m}^2, failing fold stability gates).

5. Passive Satellite Aerosol Optical Depth (AOD) Products

Passive satellite AOD retrievals (MODIS/VIIRS AOD, Sentinel-5P AOD) require clear-sky conditions and dark land surfaces. At 68.36^\circ\text{N}, persistent cloud cover, winter snow/ice surface albedo, and high solar zenith angles cause missing data rates to exceed 95\% in 2024. Furthermore, baseline aerosol loading at Abisko is pristine (\text{AOD}_{550} < 0.05), making aerosol extinction a minor contributor to shortwave forecast error compared to cloud optical variability.

Works cited

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